

RADIANT PHARMACEUTICALS LIMITED

ICT Wing

(Daily Report)

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Daily Progress Report

Day #	Date #
Day 12	June 9, 2026

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Task Assigned:

Set up the Windows Subsystem for Linux (WSL 2) and Ubuntu Linux distribution on the Windows work PC, troubleshoot virtualization/installation errors, and test the Linux environment.

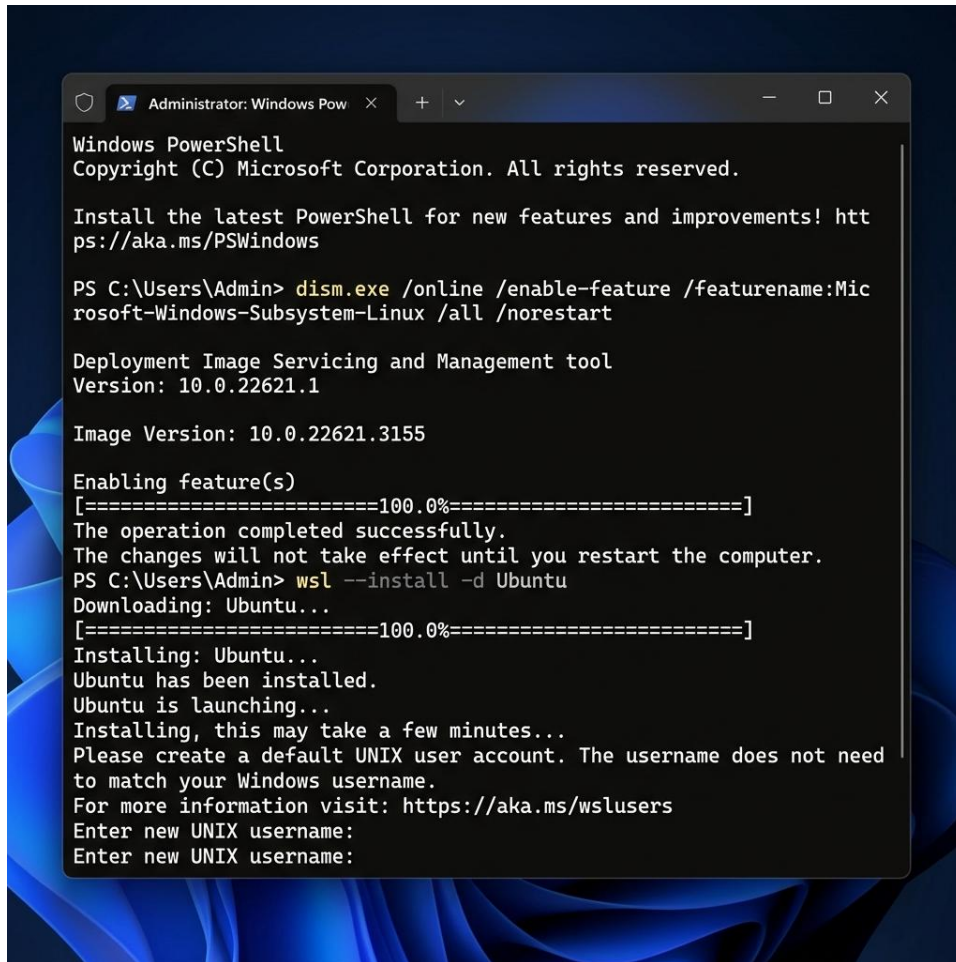
Learning Outcome:

During this task, I achieved the following learning outcomes:

- **WSL 2 Architecture & Requirements:** Learned that WSL 2 runs a real Linux kernel inside a lightweight utility VM, which offers full system call compatibility and significantly faster file system performance compared to WSL 1. Realized that WSL 2 requires hardware-level virtualization (Intel VT-x or AMD SVM) enabled in BIOS/UEFI, as well as the Windows features 'Windows Subsystem for Linux' and 'Virtual Machine Platform'.
- **WSL Installation & Configuration:** Practiced enabling system features using DISM commands in PowerShell and installing the Ubuntu distribution. Learned to initialize the Ubuntu distribution, set up a Unix user account, and run package updates/upgrades using 'sudo apt update && sudo apt upgrade'.
- **Linux Command Line & Execution:** Tested Python script execution inside the Ubuntu shell to verify package management and interpreter functionality, and checked kernel configuration details.

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Screenshots of Installation and Environment Setup:

A screenshot of a Windows PowerShell terminal window titled "Administrator: Windows PowerShell". The terminal shows the following commands and output:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Admin> dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart

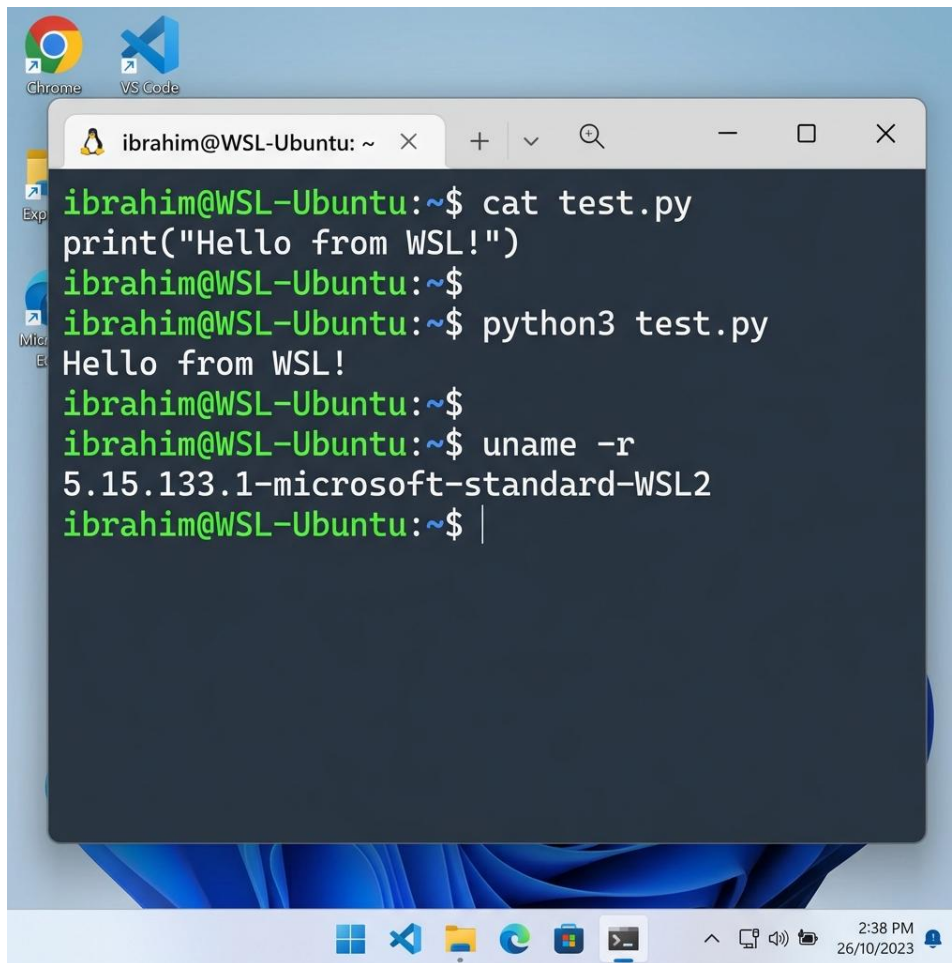
Deployment Image Servicing and Management tool
Version: 10.0.22621.1

Image Version: 10.0.22621.3155

Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.
The changes will not take effect until you restart the computer.
PS C:\Users\Admin> wsl --install -d Ubuntu
Downloading: Ubuntu...
[=====100.0%=====]
Installing: Ubuntu...
Ubuntu has been installed.
Ubuntu is launching...
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username:
Enter new UNIX username:
```

Figure 1: Enabling WSL and Virtual Machine Platform via PowerShell.

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```
ibrahim@WSL-Ubuntu: ~$ cat test.py
print("Hello from WSL!")
ibrahim@WSL-Ubuntu: ~$
ibrahim@WSL-Ubuntu: ~$ python3 test.py
Hello from WSL!
ibrahim@WSL-Ubuntu: ~$
ibrahim@WSL-Ubuntu: ~$ uname -r
5.15.133.1-microsoft-standard-WSL2
ibrahim@WSL-Ubuntu: ~$ |
```

Figure 2: Verifying Python environment and testing code execution inside Ubuntu distro.

Work Completed:

- Configured BIOS settings to enable Intel Virtualization Technology (VT-x).
- Enabled the 'Windows Subsystem for Linux' and 'Virtual Machine Platform' optional Windows features via PowerShell using DISM commands.
- Downloaded and installed the Ubuntu distribution for WSL 2.
- Set up the default Linux user account and updated the package list and upgraded installed packages using 'sudo apt update && sudo apt upgrade'.
- Created and executed a test Python script ('test.py') in Ubuntu to verify python3 runtime functionality.

Problems Faced:

- Encountered launch errors for WSL 2 and the Ubuntu distribution. The command 'wsl --install' failed to initialize because CPU virtualization was disabled in BIOS/UEFI.

GitHub Link: https://github.com/iriyasat/CSC499_Internship_2110316.git

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- The system prompted an error: 'Please enable the Virtual Machine Platform Windows feature and ensure virtualization is enabled in the BIOS.'

Solution Attempted:

- Rebooted the work PC, entered the BIOS setup utility, located CPU Configuration settings, and enabled 'Intel Virtualization Technology' (Intel VT-x).
- Opened PowerShell as Administrator and explicitly enabled the 'Virtual Machine Platform' Windows feature using:
`dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart`
- Rebooted the PC to apply the Windows feature updates, after which WSL 2 and Ubuntu initialized and launched successfully.